Cybersecurity Challenges in Digital Payments: A UPI Fraud Case Study from India

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Abstract:

With the rapid growth of digital payments in India, concerns about UPI security have increased. This study presents a case of UPI fraud in a private sector bank, focusing on how the fraud occurred, its impact on the victim, and the bank's response. Using interviews and analysis, the research highlights security gaps, lack of user awareness, and challenges in fraud detection. It also examines the financial and emotional effects on the victim and the importance of stronger security measures. The study suggests improving user awareness, strengthening fraud prevention systems, and enhancing trust in digital payment platforms.

Keywords: *UPI Fraud, Digital Payments, Cybersecurity, Online Fraud Prevention, Banking Security, User Awareness*

I. INTRODUCTION:

According to the Indian government, 84,000 cases of UPI fraud were reported in 2020–21 and 77,000 cases in 2020–21. The data was made public in the midst of an increase in online fraud cases and UPI-related cases nationwide. The UPI (Unified Payments Interface) has been tremendously increased in India. As per the National Payments Corporation of India (NPCI) the recent Nov 2023 data there are 516 participants Banks in UPI payment, and 11,235.29 (Volume (in Mn) and 17,39,740.61 (Value (in Cr.)) transactions happened in India which depicts tremendous volume of UPI transactions. The challenge to Banks, RBI and NPCI is to give secured services to the Banks customers. But many of the fraudsters are targeting to fool the Bank customers. It has been found that a lot of UPI frauds have happened and Bank customers have lost their money. The study reveals how the frauds have happened. For this study the researcher has studied the two real cases of frauds, where the researcher has personally met the victims of UPI frauds, Cyber Crime Cell, PCMC, Pune and the respective Bank in which the Victim's Bank account was. For confidentiality, here the researcher would like to pseudonym the Bank as ABC Banks, and first Victim as V1 and second Victim as V2.

II. CASE STUDY

Case 1:

Victim V1 who is well educated and very much aware about the handling of mobiles phone and their various apps. One day he received the message and call saying that your tenant has not paid the MSEB bill (Electric Bill), if not paid immediately, your MSEB connections will be disconnected. The PDF file was sent which was showing the MSEB logo on it and some text as in the image below.

Azadi Ka Amrit Mahotsav

विद्युत मंत्रालय
MINISTRY OF POWER

MAHAVITARAN

Notice for Consumer regarding to update your bill immediately

Dear consumer,
Your electricity power connection will be Disconnected tonight 09:00 pm. Because your previous month bill was not updated , please immediately call our electricity officer Mr Nitin Sharma

ministry of power would like to calrify that although we encourage our consumer's to always keep updated.

Now you update your bill by just calling our electricity helpline number .

To update your bill call our electricity helpline number :-

Call +91 9341762390

• Follow our official social media accounts @ministryofpower on twiter , facebook , instgram and YouTube

cheif electricity officer

Figure 1: pdf file received to Victim V1

The Messages received with link were as below

Figure 2: Message and link received to Victim V1

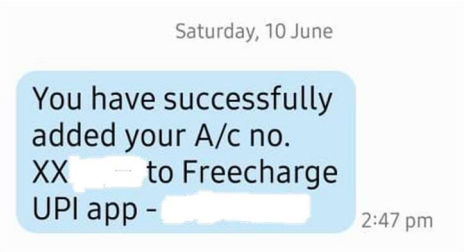

Figure 3: Message and link received to Victim V1

Later the message received to Victim V1 was "Debit INR 3.00" from the Bank account. In this case Victim V1 got alerted and he immediately transferred a big amount of money from this Bank account to another which was not connected to UPI, for the safer side. Then the Victim V1 contacted the Cyber Crime cell and discussed the issue. The Cyber cell officer instructed to register the complaint to www.Cybercrime.gov.in to file a Complaint on National Cyber Crime Reporting Portal, and told to freeze the Bank account. They were also instructed to change the UPI Pin (Google Pay) So accordingly Victim V1 tried to register a complaint to the National Cyber Crime Reporting Portal, but due complexity on the portal and its procedure of OTP each time in each login it could not happen. One problem was that OTP was receiving more delays. So, Victim V1 decided to do it later. He changed his UPI pin as instructed. After 1-2 days again another one fraudulent transaction happened from the ABC Bank account of Victim V1 of amount Rs. 11000(Dr.) and from one more his Bank account of XYZ Bank Rs. 7000 got debited without any single activity of Victim V1. It was a real shock to the Victim V1. He immediately approached both the Banks and froze his both the accounts till further instructions.

After 2-3 days the ABC Bank credited a shadow amount of Rs. 11000 to Victim V1 account. But later after Victim V1 received a message from Bank ABC, specifying the shadow amount has got debited. Crediting the shadow amount was just normal procedure by the Bank ABC as per the RBI guidelines or for the sake of reputation of the Bank. But later after 2-3 days Bank ABC replied to the Victim V1 that "you have shared the UPI id so your money could not be credited or returned by Bank". It specifies that the Bank has not taken that much effort to resolve the complaint and could return the money to which the customer (Victim V1) was

complaining.

Case 2:

In the case of Victim V2 the same case happened with the same ABC Bank as that of Victim V1. The amount of frauds that happen were only the difference. The MSEB Related link was shared by fraudsters. When the Victim V2 clicked on that link, the software like Team connect got installed on his mobile (.exe file), the Victim V2 said. Later on, the first fraudulent transactions happened (debited from his G-pay linked account) of amount Rs. 50,000 and again in 1-2 days the next fraudulent transactions happened (debited from his G-pay linked account) of amount Rs. 30000. In this case too, the Victim V2 could not file a complaint immediately on www.Cybercrime.gov.in or could not freeze his account. The Victim V2 is working in a reputed IT company.

III. INTERPRETATION

It has been observed that both the Victim V1 and V2 are educated and more familiar with handling the mobile device and were aware about the frauds type happenings recently. But even though they got trapped in the fraud and lost their money. Both of the not yet received return money from the Bank and nor any compensation for the same. Even they have not received any type of Insurance from the Bank against their lost money.

The RBI has instructed Bank customers on their RBI portal that "if someone has fraudulently withdrawn money from your bank account, inform your bank immediately." and take the acknowledgement from your bank after notifying them. The bank is supposed to resolve their complaint within 90 days from the date of receipt.

The RBI on their portal has specified "SMS on Customer Liability in Unauthorized Electronic Banking Transactions" and provided the facility for Fraudulent transactions in your bank account in which he can Limit their loss by notifying the bank immediately and specified that for more details, give a missed call on 14440.

RBI has also specified that "If the transaction has happened because of your negligence, that is, because of your sharing your password, PIN, OTP, etc., you will have to bear the loss till you report it to your bank. If the fraudulent transactions continue even after you have informed the bank, your bank will have to reimburse those amounts. If you delay the reporting, your loss will increase and it will be decided based on the RBI guidelines and the policy approved by your bank's board."

IV. RECOMMENDATIONS (Post Frauds)

For Cyber Crime Victims the recommendations are

1. Change the SIM
2. Change the UPI Pin
3. Reset the Phone or format the mobile device.
4. Aware and keep updating the types of frauds happening around and take precaution.

V. RECOMMENDATIONS (Pre Frauds)

1. Be Aware and keep updating the types of frauds happening around and take precaution.
2. Do not click any of the unknown links.
3. Keep the mobile device updated with the Operating system.
4. Use and Update the Antivirus frequently.
5. Banks, NPCI and RBI need to be aware of Bank customers in a stringent way.
6. Customers need to register their complaints immediately to www.Cybercrime.gov.in
7. Bank customers need to freeze/block their account without delay if such suspicious transactions/fraud happen, otherwise the next fraudulent transaction might happen again and again causing financial loss.

VI. CONCLUSION

The above both the cases which have happened depicts that UPI customers are not 100% safe. The mobile phones or the services given by NPCI, RBI are not that much secure or failed to protect, aware the Bank customers who are using UPI. The reply from the Bank that "your money can't be refunded as you shared your UPI pin or OTP" was very much surprising. Even after the completion of transactions the Cyber portal officials or Cyber Crime Cell can find out the fraudsters account and fraudsters where the Victim V1's and Victim V2's amount gets credited. But if it got transferred to any other account or withdrawn by fraudsters it cannot be reversed by the Bank. It has to happen before it gets transferred/withdrawn. Also, the cyber criminals are far away from the customer's location where the Cyber Crime has registered. Secondly the criminals/mastermind are different from the account holder (whom the amount gets credited). So, reaching them is cumbersome. For resolving various types of these issues, the Cybercrime cell, NPCI, RBI need to change their policy and need to secure the Bank customers with the help of advanced technology like Artificial Intelligence/Machine Learning.

Declaration of Conflicting Interests

The author declares no potential conflicts of interest with respect to the research, authorship and publication of this article.

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